

Separating Two Things Composite Modeling Provides: Knowing the Outer Function from Observing the Intermediates

1 Where this sits

Our benchmark study compared two surrogates on problems of the composite form $f(x) = g(h(x))$, where h is an expensive black box and g is a known, cheap function. *Direct* fits a Gaussian process to f ; *Composite* fits Gaussian processes to h and pushes their posterior samples through g . Composite improved most comparisons, but by amounts ranging from 121% to nothing, and the follow-up experiments are designed to find out what separates those cases.

This note addresses a confound that runs through all of them. Composite gives the surrogate **two different things at the same time**:

1. the **knowledge of g** — the surrogate is told the algebraic form of the final objective, so it never has to learn that part from data;
2. the **observations of h** — each expensive evaluation returns k numbers instead of one.

Every Direct–Composite comparison we have run, and every experiment currently planned, changes both at once. So when Composite wins by 121% on DTLZ2 we cannot say which of the two is responsible, and when it wins by nothing on RGB we cannot say which one failed to pay off.

The fix is to add a third surrogate that receives exactly one of the two.

2 Three arms

Fix a single objective $f(x) = g(h(x))$ with $h : \mathcal{X} \rightarrow \mathbb{R}^k$ unknown and $g : \mathbb{R}^k \rightarrow \mathbb{R}$ known. Write $\mathcal{D}_n = \{(x_i, \cdot)\}_{i \leq n}$ for the data after n evaluations.

Arm	Observes	Knows g	Surrogate
1. Direct	$y_i = f(x_i)$	no	$f \sim \mathcal{GP}$ fit to y
2. Composite	$h(x_i)$	yes	$h_j \sim \mathcal{GP}$ fit to h_j , then apply g
3. Latent	$y_i = f(x_i)$	yes	$h_j \sim \mathcal{GP}$ <i>a priori</i> , h never observed

Arm 3 is the new one. It has the same prior belief about h that Composite has, and it knows g exactly, but it only ever sees the scalar y . Its posterior over the latent values $\mathbf{h}(X) = (h(x_1), \dots, h(x_n))$ is just Bayes’ rule,

$$p(\mathbf{h}(X) \mid \mathbf{y}) \propto \underbrace{\prod_{i=1}^n \mathcal{N}(y_i; g(h(x_i)), \sigma^2)}_{\text{likelihood}} \cdot \underbrace{\prod_{j=1}^k \mathcal{N}(h_j(X); m_j, v_j K)}_{\text{GP prior}}, \quad (1)$$

and prediction at a new point is exact given the latent values, because $h(x') \mid \mathbf{h}(X)$ is an ordinary GP conditional. The arm-3 predictive is therefore a *mixture of Gaussian processes*, one component per posterior sample of $\mathbf{h}(X)$.

Because arms 2 and 3 use the **identical** prior on h — the true kernel of the generator that produced the test functions — the only difference between them is what they observe. The two comparisons then read cleanly:

$$\text{Arm 3} - \text{Arm 1} = \text{the value of } \textit{knowing } g, \quad (2)$$

$$\text{Arm 2} - \text{Arm 3} = \text{the value of } \textit{observing } h. \quad (3)$$

3 Why the decomposition should be informative

The second quantity has a clean characterization: observing h can only help to the extent that $y = g(h)$ fails to determine h . That makes the *invertibility of g* the governing property, and it gives three regimes we can reason about in advance.

3.1 Invertible g : observing h is worth nothing

If g is injective and the observations are noise-free, then $h(x_i) = g^{-1}(y_i)$ exactly. Arm 3 recovers the intermediates from the objective alone, so arms 2 and 3 condition on identical information and are the *same model*. Any advantage Composite shows over Direct on such a problem is attributable entirely to knowing g .

The doc’s exponential warping family, $g(z) = \exp(\beta z)$, is of this type. Knowing g is still worth something — it tells the surrogate that $y > 0$ and how the scale is warped — but the extra numbers are redundant.

3.2 Linear g : knowing g is worth nothing

If $g(\mathbf{y}) = \sum_j c_j y_j$ and the h_j are independent GPs, then $g(h(\cdot))$ is *itself* a GP, with mean $\sum_j c_j m_j$ and kernel $(\sum_j c_j^2 v_j)k(\cdot, \cdot)$. Arm 3 therefore collapses to an ordinary GP fitted to y — which is arm 1 with a particular kernel. Up to hyperparameter estimation, knowing g buys nothing at all.

This is the sharpest prediction in the design, and it matters for the paper: Truss and RCM40 both have linear outer maps, and both produced our weakest results. If the prediction holds, the explanation is not merely that “linear maps are easy” but something more specific — on a linear map there is no knowledge to exploit, so whatever gain Composite shows must come from the extra observations alone.

3.3 Non-invertible g : both channels are live

Take the smallest interesting case, $g(z) = z^2$, with a noise-free observation $y = g(h(x))$. Then $h(x)$ is known only up to sign,

$$p(h(x') \mid g(h(x)) = y) = p(h(x') \mid h(x) \in \{+\sqrt{y}, -\sqrt{y}\}), \quad (4)$$

so the posterior is a two-component mixture for one observation, and a mixture over 2^n sign patterns for n of them, weighted by how plausible each pattern is under the GP prior. Observing h removes that ambiguity outright, which is what arm 2 measures.

But knowing g is worth something even without any observation of h . The range of g is $[0, \infty)$, so arm 3 assigns

$$P(y(x') = -1 \mid \mathcal{D}) = P(h(x')^2 = -1 \mid \mathcal{D}) = 0, \quad (5)$$

whereas a Direct GP on y — a Gaussian — puts positive probability on negative values it can never observe. More generally, knowing g imposes the support, the shape, and any kinks or saturation of the outer map, none of which a stationary GP on y can represent.

In general the information destroyed by g is the *fiber* $g^{-1}(y)$: a finite set for z^2 or $\cos$, a half-line for a clipped or rectified map, a circle for a Euclidean norm, a hyperbola for a product. This is the structure the synthetic maps in Section 4 are chosen to vary.

3.4 Summary of predictions

g	Type	Arm 3 – Arm 1 (knowing g)	Arm 2 – Arm 3 (observing h)
z	identity	0	0
$\exp(\beta z), \sinh(\beta z)$	invertible	+	0 by construction
$(z - c)^2$	lossy, tunable	+	+ at $c = 0$, $\rightarrow 0$ as c grows
$ z $	lossy	+	+
$y_1 + y_2$	linear	≈ 0	+
norm, max, product	lossy	+	+

4 Test functions

4.1 Design principle

The intermediates must not become a confound. Every primitive h_j is therefore a fixed draw from the same RBF-GP family used by the surrogate, generated once with random Fourier features,

$$h(x) = \sqrt{\frac{2}{M}} \sum_{m=1}^M a_m \cos(\omega_m^\top x + b_m), \quad a_m \sim \mathcal{N}(0, 1), \quad b_m \sim U(0, 2\pi), \quad \omega_m \sim \mathcal{N}(0, \ell^{-2}I), \quad (6)$$

with $M = 1024$ and $\ell = 0.5$, which has prior $\mathcal{GP}(0, k_{\text{RBF}})$. The draw is standardized on a fixed Sobol reference set, so the implied prior on the standardized function is $\mathcal{GP}(-\mu/s, k_{\text{RBF}}/s^2)$; those constants are what arms 2 and 3 use as their known prior. The same draw seeds are reused for every condition, so switching g never changes the underlying functions.

Standardization matters for a reason beyond scale: it puts the draw’s mean at zero, so h takes *both signs*. Without that, z^2 would be injective over the region actually visited and would lose nothing.

Each final objective is normalized to roughly $[0, 1]$ using the 1st and 99th percentiles of $g(h(\cdot))$ on a fixed reference set, so that different outer maps do not differ merely in scale. The normalization is affine and known, so it is part of g for all three arms.

4.2 The maps

The sweep. $g_c(z) = (z - c)^2$ for $c \in \{0, 0.5, 1, 2, 3\}$ holds the functional form fixed and varies only how much ambiguity there is. At $c = 0$ the sign of h is lost wherever the draw crosses zero; at $c = 3$ the shifted argument almost never changes sign over the domain, so g is effectively injective. This is the cleanest available knob on “how much g discards.”

Matched pairs. Each pair changes one property and holds the rest:

Pair	Shared	Isolates
z^2 vs. $ z $	both two-to-one	smoothness at the optimum
$\exp(\beta z)$ vs. $\sinh(\beta z)$	both injective, both smooth	whether a range constraint helps
$\cos(\omega z)$, $\omega \in \{1, 2, 4\}$	smooth	<i>how many</i> preimages
$\max(z, 0)$, $\text{clip}(cz, -1, 1)$	—	censoring and saturation

Benchmark analogs. Two intermediates into one objective, each mirroring a real benchmark: $y_1 + y_2$ (Truss, RCM40), $y_1 y_2$ (DTLZ2), $\sqrt{y_1^2 + y_2^2}$ (Color), $\max(y_1, y_2)$ (Lake), and $u_1/(0.05 + u_1 + \lambda u_2)$ with $u = \text{sigmoid}(y)$ (RGB). These let the synthetic conclusions be carried back to the benchmark results.

5 Inference for the latent arm

Arm 3 is the only one that needs more than a closed form. Three cases:

Closed form where possible. For the identity and injective maps, $h = g^{-1}(y)$ exactly, so arm 3 is implemented as arm 2. For a linear map the posterior (1) is Gaussian and is available analytically. Both shortcuts are validated against the general sampler in Section 8.

General case. Otherwise we sample (1) over the $n \times k$ latent values, which is at most 90 numbers at our budget. Two move types alternate.

Elliptical slice sampling [1] is the standard sampler for a GP prior with an arbitrary likelihood: it proposes $\mathbf{h} \cos \theta + \boldsymbol{\nu} \sin \theta$ with $\boldsymbol{\nu}$ a prior draw, and shrinks θ until the likelihood threshold is met. It needs no gradients, which matters because $|z|$, $\max$ and the clipped map have kinks or flat regions, and no tuning.

Level-set moves do the work that elliptical slice sampling cannot. When σ is small the likelihood is sharply peaked on the set $\{g(h(x_i)) = y_i\}$, and slice proposals must shrink to roughly $\sigma/\text{sd}(h)$ to stay on it, so movement *along* that set is a slow random walk — precisely along the directions the experiment is about. But those directions are known in closed form for each g . We therefore propose $h(x_i) \mapsto h'(x_i)$ with

$$g(h'(x_i)) = g(h(x_i)) \text{ exactly,} \quad (7)$$

from a symmetric proposal, so the likelihood ratio is identically one and the Metropolis–Hastings acceptance probability reduces to the GP prior ratio alone. With $Q = K^{-1}$ and $r_j = h_j(X) - m_j$, changing coordinate i of latent j from α to β gives

$$\Delta \log p = -\frac{1}{2v_j} \left[Q_{ii}(\beta^2 - \alpha^2) + 2(\beta - \alpha)((Qr_j)_i - Q_{ii}\alpha) \right]. \quad (8)$$

The moves are the symmetries of each map: reflection $h \mapsto 2c - h$ for $(z - c)^2$ and $|z|$; reflection through the nearest multiple of π/ω plus translation by the period for $\cos(\omega z)$; a random walk confined to the flat region for $\max(z, 0)$ and the clipped map; rotation of (h_1, h_2) for the norm; negation and reciprocal rescaling for the product; swapping the two coordinates and moving the smaller one for the max; and the shift $(h_1 + \delta, h_2 - \delta)$ for the sum. Only the ratio map has no convenient exact symmetry, and is sampled by elliptical slice sampling alone.

Practical details. Burn-in anneals σ geometrically down to its target so that chains started from the prior can find the data; later fits warm-start from the previous chain, with the new point initialized from its GP conditional. Several chains are run and diagnostics are computed on the predictive of f , which is invariant to the symmetries of h that leave f unchanged — so a chain that has globally flipped sign under an even g does not register as non-convergence.

6 What is measured

Information destroyed by g . The average posterior variance of the intermediates *at the points already evaluated*,

$$U_{\text{lost}} = \frac{1}{nk} \sum_{i=1}^n \sum_{j=1}^k \text{Var}(h_j(x_i) \mid \mathbf{y}), \quad (9)$$

is zero when g is injective and grows with the size of the fiber. It is a measured version of the quantity the whole design is organized around, and the central plot is the “observing h ” gain against it.

Optimization performance. Each arm reports simple regret against a reference minimum; comparisons use the final $\log_{10}$ regret, paired over seeds, since all three arms share an initial design and an acquisition procedure.

A confound to control. Arms 2 and 3 are given the generator’s true kernel, whereas arm 1 must estimate its hyperparameters by maximum likelihood, because for a general g there is no true kernel for f . The “knowing g ” comparison therefore mixes the value of g with the value of a correct prior. The identity map measures the second effect alone — knowing g is worth exactly nothing there — and so serves as the baseline to subtract. A robustness run with maximum-likelihood hyperparameters in all three arms resolves it directly.

7 Protocol

Single objective, $d = 6$, $\ell = 0.5$; $n_0 = 5$ Sobol points and 40 adaptive evaluations, matching Experiment 1; 10 paired seeds. All three arms share the initial design, the acquisition function (Monte Carlo LogEI), the candidate pool and the evaluation budget, so the surrogate is the only thing that differs.

8 Validation

The sampler is checked against the two cases whose answers are known independently before being trusted anywhere else.

Linear map, against the exact posterior. On $y_1 + y_2$ with $n = 20$, the sampled posterior matches the analytic Gaussian: the error in the posterior mean is a median of 0.065 posterior standard deviations (max 0.249), and the ratio of sampled to exact standard deviation has median 0.973 and range $[0.85, 1.10]$.

Injective map, for calibration. On $\exp(z)$ the true h falls within two posterior standard deviations at 95% of points. The largest raw error, 0.62, occurs at $h = -1.96$, where $\exp$ is flat enough that its slope falls below the likelihood noise. This is correct behavior and a substantive caveat: **with observation noise, an injective g still destroys information wherever it is flat**, so “arm 2 equals arm 3 for injective g ” holds exactly only in the noise-free limit.

The sweep behaves as designed.

Map	U_{lost}	Chain disagreement	Move acceptance
$(z - 0)^2$	0.877	0.029	0.67
$(z - 0.5)^2$	1.072	0.052	0.52
$(z - 1)^2$	0.443	0.047	0.32
$(z - 2)^2$	0.139	0.036	0.07
$(z - 3)^2$	0.003	0.018	0.00
$ z $	0.882	0.027	0.66

The information destroyed falls by more than two orders of magnitude across the sweep, and the level-set acceptance rate falls with it — at $c = 3$ the reflected proposal is so implausible under the prior that it is never accepted, which is the correct behavior when there is no ambiguity left to explore. Chain disagreement is in normalized objective units, where the objective spans about one.

9 Limitations

- **Noise makes injectivity a matter of degree.** As measured above, an injective g still loses information where its slope is small relative to σ . The clean statement holds only in the noise-free limit.
- **Continuous fibers mix more slowly than discrete ones.** Reflections for z^2 are single moves that jump between modes; following the circle fiber of a norm is a single-site random walk over a smooth field, which mixes more slowly when observed points are strongly coupled. The ratio map has no exact moves at all and relies on elliptical slice sampling.
- **The reference minimum is estimated**, from a large Sobol set refined by gradient ascent, so regret is measured against a very good estimate rather than a proven optimum.
- **Single objective.** The decomposition is cleanest with one objective; whether the same accounting holds once a hypervolume acquisition is involved is a separate question, and the natural next step once these results are in.

References

- [1] I. Murray, R. P. Adams, and D. J. C. MacKay. Elliptical slice sampling. In *Proceedings of the 13th International Conference on Artificial Intelligence and Statistics*, PMLR 9:541–548, 2010.